

Heart Disease Analysis Project

1. Introduction

The purpose of this project is to analyze sales data using Tableau for interactive visualization, MySQL for database management, and Flask for deploying the dashboard through a web application.

The project demonstrates the complete data analytics lifecycle from data collection to visualization and deployment.

2. Objectives

- Download a sales dataset.
 - Store the dataset in MySQL Database.
 - Perform SQL operations for analysis.
 - Connect Tableau with MySQL.
 - Clean and prepare the data.
 - Create interactive dashboards.
 - Publish Tableau Dashboard.
 - Embed the dashboard into a Flask application.
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3. Technologies Used

Tool	Purpose
Python	Data Processing
Pandas	Data Cleaning
MySQL	Database Storage
MySQL Workbench	SQL Queries
Tableau Desktop	Visualization

Tool	Purpose
Tableau Public/Cloud Publishing Dashboard	
Flask	Web Application

4. Project Workflow

Dataset



Data Cleaning



Tableau Connection



Data Preparation



Visualizations



Dashboard



Publish Dashboard



story



Performance testing



Web Integration



Documentation

5. Downloading the Dataset

Heart Disease Dataset



Dataset Contains

- Age
- Gender
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol
- Fasting Blood Sugar
- Resting ECG
- Maximum Heart Rate
- Exercise-Induced Angina
- ST Depression
- Slope
- Number of Major Vessels
- Thalassemia
- Heart Disease Status (Target)

6. Storing Data in MySQL Database

Create Database

```
CREATE DATABASE heart_disease_db;
```

Select Database

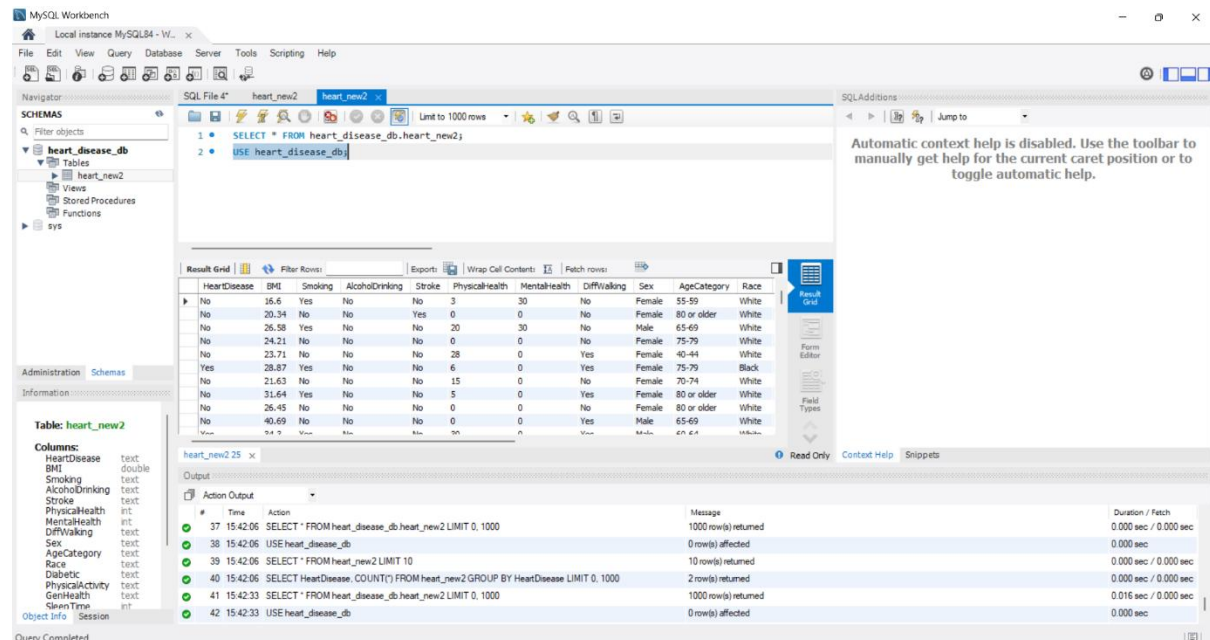
```
USE heart_disease_db;
```

Create Table

CREATE TABLE heart_disease (

Import Dataset

The CSV dataset was imported into MySQL using MySQL Workbench's Table Data Import Wizard.



SQL Operations Performed

Total Patients

```
SELECT COUNT(*) FROM heart_disease;
```

Patients with Heart Disease

```
SELECT COUNT(*)
```

```
FROM heart_disease
```

```
WHERE target = 1;
```

Patients without Heart Disease

```
SELECT COUNT(*)
```

```
FROM heart_disease
```

```
WHERE target = 0;
```

Average Cholesterol

```
SELECT AVG(chol)
```

```
FROM heart_disease;
```

Average Maximum Heart Rate

```
SELECT AVG(thalach)
FROM heart_disease;
```

Gender-wise Heart Disease

```
SELECT sex,
COUNT(*)
FROM heart_disease
WHERE target=1
GROUP BY sex;
```

Chest Pain Type Distribution

```
SELECT cp,
COUNT(*)
FROM heart_disease
GROUP BY cp;
```

Connecting MySQL with Tableau

The MySQL database was connected to Tableau using the MySQL Connector/J driver.

Connection Details

- Server: localhost
- Port: 3306
- Database: heart_disease_db
- Username: root

After successful authentication, the dataset was imported into Tableau.

Tableau - Book1

File Data Server Window Help

Connections: localhost

Database: heart_disease_db

Table: heart_new2

heart_new2 (heart_disease_db)

Connection: Live Extract

Filters: 0 Add

heart_new2

18 fields 4500 rows

100 rows

Name: heart_new2

Description: No description available.

Fields:

Type	Field Name	Phys...	Rem...
Abc	Heart Disease	heart_...	Heart...
#	BMI	heart_...	BMI
Abc	Smoking	heart_...	Smoki...
Abc	Alcohol Drinking	heart_...	Alcoh...
Abc	Stroke	heart_...	Stroke
#	Physical Health	heart_...	Physic...
#	Mental Health	heart_...	Menta...
Abc	Diff Walking	heart_...	DiffWa...

Sex	Age Category	Race	Diabetic	Physical Activity	Gen Health	Sleep Time
Female	55-59	White	Yes	Yes	Very good	
Female	80 or older	White	No	Yes	Very good	
Male	65-69	White	Yes	Yes	Fair	
Female	75-79	White	No	No	Good	
Female	40-44	White	No	Yes	Very good	
Female	75-79	Black	No	No	Fair	
Female	70-74	White	No	Yes	Fair	
Female	80 or older	White	Yes	No	Good	
Female	80 or older	White	No, borderline diabetes	No	Fair	
Male	65-69	White	No	Yes	Good	
Male	60-64	White	Yes	No	Poor	

Go to Worksheet

Data Source Sheet 1

9. Data Preparation

The following data preparation tasks were performed:

- Checked for missing values.
- Verified data types.
- Renamed columns for readability.
- Removed duplicate records.
- Created calculated fields.
- Validated data consistency.

MySQL Workbench

Local instance MySQL84 - MySQL

File Edit View Query Database Server Tools Scripting Help

SCHEMAS: heart_disease_db

Tables: heart_new2

Columns: 18

Indexes: 4337

Foreign Keys: 367

Triggers: 1.5 MB

Views: 0.0 bytes

Stored Procedures: 0.0 bytes

Functions: 1.5 MB

Information: 2024-07-08 15:44:49

Table Details: heart_disease_db.heart_new2

Engine: InnoDB

Row format: Dynamic

Column count: 18

Table rows: 4337

Table row length: 367

Data length: 1.5 MB

Index length: 0.0 bytes

Max data length: 0.0 bytes

Data free: 0.0 bytes

Table size (estimated): 1.5 MB

Update time: 2024-07-08 15:44:49

Create time: 2024-07-08 15:44:49

Auto increment: 0

Information on this page may be outdated. Click [Analyze Table](#) to update it.

Query Output:

Time	Action	Message	Duration / Fetch
115	16:39:06	USE heart_disease_db	0 rows affected / 0.000 sec
116	16:39:06	SELECT * FROM heart_new2 LIMIT 10	10 rows returned / 0.000 sec / 0.000 sec
117	16:39:06	SELECT * FROM heart_new2 GROUP BY Sex LIMIT 0, 1000	2 rows returned / 0.016 sec / 0.000 sec
118	16:39:06	SELECT AVG(BMI) FROM heart_new2 LIMIT 0, 1000	1 row returned / 0.000 sec / 0.000 sec
119	16:39:06	SELECT Smoking, COUNTY FROM heart_new2 GROUP BY Smoking LIMIT 0, 1000	2 rows returned / 0.000 sec / 0.000 sec
120	16:39:06	USE heart_disease_db	0 rows affected / 0.000 sec

Object Info Session

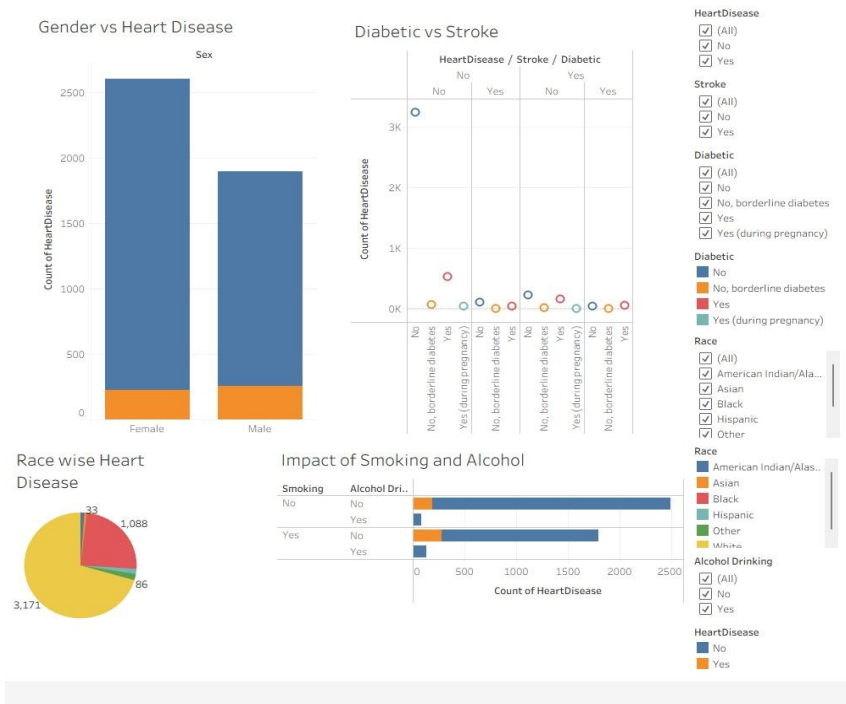
Query Completed

Dashboard Development

The dashboard was developed to provide insights into heart disease patterns.

Dashboard Features

Heart Disease Analysis Dashboard



The dashboard includes **8 unique visualizations**.

Gender wise Heart Disease

Age wise Heart Disease

People Suffering Diabetic and Stroke

Impact of Smoking and alcohol drinking on heart disease

Other Diseases vs Stroke

Race wise Heart disease

General Health vs Heart Disease

Physical activity vs heart disease

Age and BMI vs Heart disease

People got stroke suffering from Diabetes and Heart disease

Dashboard Filters Used

The dashboard includes **6 interactive filters**.

- Age Group
 - Gender
 - Chest Pain Type
 - Cholesterol Range
 - Heart Disease Status
 - Exercise-Induced Angina
-

Calculated Fields

BMI category

Risk Category

IF [Age] >= 60 THEN

"High Risk"

ELSEIF [Age] >= 40 THEN

"Medium Risk"

ELSE

"Low Risk"

END

The screenshot shows the Tableau Desktop interface with the following components:

- Data Source:** heart_new2 (heart_disea...)
- Columns Shelf:** BMI category
- Rows Shelf:** (Empty)
- Filters Shelf:** (Empty)
- Marks Shelf:** SUM(BMI) (Green pill)
- Table List:** Includes tables like AgeCategory, Alcohol Drinking, Asthma, BMI category, Diabetic, Diff Walking, Gen Health, Heart Disease, Kidney Disease, Physical Activity, Race, Risk Group Classification, Sex, Skin Cancer, Smoking, Stroke, and Measure Names.
- Sheet 1:** A table with columns BMI category and Risk Group ..

BMI category	Risk Group ..
Normal	Normal
Obese	Obese
Overweight	Overweight
Underweight	Normal

Publishing Dashboard

The Tableau dashboard was published to **Tableau Public**.

The published Tableau dashboard was embedded into a Flask web application using an HTML iframe.

Project Outcomes

The project successfully achieved the following objectives:

- Downloaded the Heart Disease dataset.
- Stored the dataset in MySQL.
- Performed SQL analysis.
- Connected MySQL with Tableau.
- Prepared and cleaned the data.
- Created **8 interactive visualizations**.
- Implemented **6 dashboard filters**.
- Created **4 calculated fields**.
- Published the Tableau dashboard.
- Embedded the dashboard into a Flask web application.

Conclusion

This project demonstrates a complete Business Intelligence workflow using **MySQL**, **Tableau**, and **Flask** for heart disease analysis. The dataset was stored in a relational database, analyzed using SQL queries, visualized through interactive dashboards, and deployed as a web application. The dashboard provides valuable insights into patient demographics, medical attributes, and heart disease risk factors, enabling users to explore patterns and support informed healthcare decisions. The project showcases practical skills in data management, SQL analysis, dashboard design, data visualization, and web deployment.